MTH 207 Practice Exam 2

First Name	Last Name	Email
		@uwlax.edu

Rules

- The cover page should only have your name and email. Nothing else may may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating "does not exist" without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

1. Find $\lim_{x \to 0} \frac{\sin(3x)}{4x}$

2. Find $\lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$

3. Find $\lim_{x \to 3} \frac{x-3}{x^2-9}$

4. If $f(x) = \dfrac{x-3}{x^2-9}$, find $f'(x)$

5. If $y = \sqrt{x^9 - 10x^7 + 5^x}$ find $\dfrac{dy}{dx}$

6. If $f(x) = x^3 \cdot 6^x + 5\cos(x) + \sin\left(\sqrt[3]{\sin(x)}\right)$, find $f'(x)$

7. If $y = 3x^5 \cdot 7^{\sin(x)+\cot(x)}$ find $\dfrac{dy}{dx}$

8. If $f(x) = 4\tan(x) + 6^x + \dfrac{x^3 + x^2 + x + 1}{x}$, find $f''(x)$

9. If $y = \dfrac{3 + x^9 \sin(4x)}{2^x + x^9} + \sin\left(\dfrac{x^2}{x^4 + 1}\right)$ find $\dfrac{dy}{dx}$

10. If $f(x) = 0.1x^2 + x$ describes the position (in meters) of Brewers coach Pat Murphy's broom x minutes after $12:00$ noon, what is the velocity of Murphy's sweeping 10 minutes after noon?

11. If $f(x) = x^2 + \sin(x)$, find $f'(\frac{4\pi}{3})$ and **SIMPLIFY**.

12. If y depends on x according to $x^3 + y^5 = x + \sin(x^7 y^9)$, find $\frac{dy}{dx}$.

13. Find an equation for the tangent line to $f(x) = 2x^4$ at $x = 10$.

14. Starting from $y = \mathsf{sec}(x)$ find a formula for $\frac{dy}{dx}$, showing all work including a relevant trig identity.

15. Starting from $y = \mathsf{arccos}(x)$ find a formula for $\frac{dy}{dx}$, showing all work including a relevant trig identity.